

Module 05: Action — Functions, Loops, and Making Things Happen

Learning Objectives

1. Use **functions** to organize actions into reusable, named blocks of code.
2. Use **loops** (for, while) to repeat actions efficiently.
3. Connect functions and loops to the Active Inference sense-decide-act cycle.

Introduction

Action is the part of Active Inference where the agent *does* something — it changes the world. In code, actions are functions and loops: the building blocks that make things happen. This module teaches you to write better functions and loops, and to see them as the "motor system" of your code agents.

Key Concepts

1. Functions as Actions

A function is a named action. Instead of writing the same code over and over, you write it once and *call* it:

```
def move_forward(player, distance):  
    player.x += distance  
    print(f"Moved to {player.x}")  
  
def turn_left(player):  
    player.direction = rotate(player.direction, -90)  
    print(f"Facing {player.direction}")
```

Each function is a discrete **action** your agent can take. The function name is the action's label. The parameters customize the action.

2. Loops as Repeated Actions

A `while` loop is an action repeated until a condition is met. This is the core of the Active Inference loop:

```
while not at_goal:
    observation = sense(world, player)      # Sense
    decision = decide(observation, player)  # Decide
    at_goal = act(decision, player, world)  # Act
```

The loop keeps running — sense, decide, act — until the agent reaches its goal. Every game, every robot, every AI runs a loop like this.

3. Side Effects: Actions Change the World

A function that only calculates something (like `add(3, 4)`) is pure computation. But a function that changes a variable, moves a character, or writes to a file has a **side effect** — it changes the world. In Active Inference, every action is a side effect: the agent acts on the environment to make it match its predictions.

Activities

Activity 1: Action Library

Create a library of at least 5 functions for a game character: `move_forward()`, `turn_left()`, `turn_right()`, `pick_up(item)`, `drop(item)`. Write a main loop that lets the user type commands ("forward", "left", "pick up key") and calls the right function.

Activity 2: The 100-Step Challenge

Write a program where a bot must reach a treasure in a 20×20 grid. The bot can move forward, turn left, or turn right. Challenge: can your bot reach the treasure in fewer than 100 steps? Optimize your action loop!

Summary

Functions are named actions. Loops repeat the sense-decide-act cycle. Side effects are how code agents change the world. Well-organized functions and efficient loops are the "muscles" of your code — the tools that turn decisions into results.

References

- Matthes, E. (2019). *Python Crash Course*. Chapters 8-9.